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FAILURE OF WAVE THEORY

Three major features of the photoelectric effect cannot be explained in terms of the wave theory of light: →

1. The maximum kinetic energy should be proportional to the intensity of radiation. According to wave theory the kinetic energy of the photoelectrons should increase as the light beam is made more intense.

Light is an electromagnetic wave. When an electromagnetic wave is incident on the surface, under the influence of the alternating electric field of the incident light wave, the electrons in the target start oscillating back & forth. Under certain conditions, the oscillating electron will pick up enough energy to break through the surface of the target. If the intensity of the beam is increased, the amplitude of the electric field is increased and it seems reasonable to suppose that this stronger alternating electric field will give a more energetic kick to the ejected electron. Equivalently, increasing the intensity of the light source, increases the electric field $\vec{E}$ of the wave which also increases the force $\vec{F} = -e\vec{E}$ on the electron and its kinetic energy when it eventually leaves the surface. However, fig(3) shows that $K_{\max}$ ($= eV_0$) is independent of the light intensity.

It is observed that for a given frequency the intense & the feeble light beams give exactly the same maximum kick to breakaway electrons.

2. The photoelectric effect should for light of any frequency a wavelength. according to wave theory. Since light is an electromagnetic wave, According to wave theory, as long as the light is intense enough to release electrons, the photoelectric effect should occur no matter what the frequency or wavelength. There should not be any threshold wavelength. Experiments, however, show the existence of threshold wavelength.

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3. The first electrons should be emitted in a time interval of the order of seconds after the radiation first strikes the surface. In the wave theory, the energy of the wave is uniformly distributed over the wavefront. If the electron absorbs energy directly from the wave, the amount of energy delivered to any electron is determined by how much radiant energy is incident on the surface area in which the electron is confined. Assuming this area is about the size of an atom, a rough calculation leads to an estimate that the time lag between turning on the light and observing the first photoelectrons should be of the order of seconds. For example let us consider a laser beam with an intensity of 120 W/m^2 is incident on a surface of sodium. It takes a minimum energy of 2.3 eV to release an electron from sodium. Assuming the electron to be confined to an area of radius equal to that sodium atom ($0.10 \text{ nm} = 1 \times 10^{-10} \text{ m}$), how long will it take for the radius surface to absorb enough energy to release an electron?

$$\begin{aligned} \text{Here intensity } I &= 120 \text{ W/m}^2 \\ \text{electron is confined to area} &= \pi r^2 \\ &= 3.1 \times (10^{-10})^2 \text{ m}^2 \\ &= 3.1 \times 10^{-20} \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Average Power delivered by the wave (P}_{\text{av}}) &= IA \text{ watt} \\ &= 3.1 \times 10^{-20} \times 120 \text{ W} \end{aligned}$$

$$\text{Now } P_{\text{av}} = \frac{\Delta E}{\Delta t}$$

$$3.1 \times 10^{-20} \times 120 \text{ W} = \frac{2.3 \times 1.6 \times 10^{-19} \text{ J}}{\Delta t}$$

$$\Delta t = \frac{2.3 \times 1.6 \times 10^{-19} \text{ J}}{3.1 \times 10^{-20} \times 120 \text{ W}}$$

$$= 0.10 \text{ s.}$$

In reality, electrons in metals are not always bound to individual atoms but instead are free

to room throughout the metal. However, no matter what reasonable estimate we make for the area over which the energy is absorbed, the characteristic time for photoelectron emission is estimated to have a magnitude of the order of seconds, in a range easily accessible to measurement.

Another example: $\rightarrow$ Let us consider violet light falling on sodium metal in the above apparatus experimental apparatus. There will be a detectable photoelectric current when 10^{-6} W/m^2 of em energy is absorbed by the surface. Now there are 10^{19} atoms in a layer of sodium one atom thick and 1 m^2 in area. If it is assumed that the incident light is absorbed in the 10 upper most layer of sodium atoms, the energy 10^{-6} W/m^2 is distributed among 10^{20} atoms.

$$\text{Hence each atom receives average energy power } P_{av} = \frac{10^{-6} \text{ W}}{10^{20}}$$

$$= 10^{-26} \text{ W} = 10^{-26} \text{ J/s}$$

Therefore, for any single electron to accumulate 1 eV energy for emission

$$t = \frac{\Delta E}{P_{av}}$$

$$= \frac{1 \text{ eV}}{10^{-26} \text{ W}}$$

$$= \frac{1 \times 1.6 \times 10^{-19} \text{ J}}{10^{-26} \times \text{J/s}}$$

$$= 1.6 \times 10^7 \text{ s}$$

But it is experimentally found that whether the light is intense or feeble, the photoelectric emission is instantaneous.

EINSTEIN PHOTON THEORY

A successful theory of the photoelectric effect was developed by Albert Einstein in 1905. He based his arguments on Planck's idea of quantum theory. Five years earlier, in 1900, the German physicist Max Planck developed a theory to explain the spectrum of black body radiation. Planck assumed that the radiation is emitted discontinuously as little bundles/packets/bursts of energy. The packets of energy are called quanta or photons. Planck found that light-the quanta associated with a particular frequency ν of light all have the same energy & this energy E is directly proportional to ν . That is

$$E = h\nu$$

where h is a constant called Planck's constant & its value

$$h = 6.63 \times 10^{-34} \text{ Joule-sec.}$$

While Planck assumed that light although emitted from its source discontinuously, but it propagates continuously through space as electromagnetic waves.

Einstein postulated that light is not only emitted in quanta but also travels as quanta i.e. the photons preserve their identity throughout their life.

Properties of photons

- A photon always travels at a speed $c = 3 \times 10^8 \text{ m/s}$ in vacuum. This is true for any frame of reference used to observe the photon.
- The rest mass of photon is zero.
- Each photon has a definite energy & a definite linear momentum

$$E = h\nu$$

$$p = \frac{E}{c} = \frac{h}{\lambda}$$

Thus, all photons of light of a particular wavelength have the same energy & the same momentum.

- If the intensity of a given frequency is increased

there is an increase in the no. of photons crossing a given area in a given time. The energy of each photon remains the same.

(e) A photon may collide with a material particle. Total energy and the total momentum remain conserved in such a collision. The photon may get absorbed and/or a new photon may be emitted. Thus, the number of photons may not be conserved.

Explanation: →

Einstein interpreted a photoelectron is released as a result of an encounter with a single photon. When a photon collides with an electron of the metal, it transfers its energy to the electron a single photoelectron instantaneously. This is an 'all' or 'none' process i.e. either the photon gives whole of its energy $h\nu$ to the electron or no energy at all. When a photon of energy $h\nu$ is incident on a metal surface, its energy is used up in two ways: -

1. A part of energy of the photon is used in ejecting out the electron from the binding forces of the nucleus. This energy used in doing work to eject out of electron from the atom is expressed as the work function of the photo-metal.

Work function : → It is defined as the minimum amount of energy required to liberate electron from the bondage of its atom. It is denoted by ϕ .

2. The remaining energy ($h\nu - \phi$) of the photon is used to impart kinetic energy to the same electron.

It is clear that the energy of the photon is not divided; but the whole energy of a photon is used by the same electron in ejecting it out from the parent atom and imparting it a velocity.

It is also clear that if the electron

does not lose energy by internal collisions, as it escapes from the metal, all the remaining energy $(h\nu - \phi)$ appears as kinetic energy of the electron. Thus $h\nu - \phi$ represents maximum kinetic energy ($K_{\max}$).

If the electron makes some collisions before coming out, the kinetic energy will be less than $(h\nu - \phi)$. The actual kinetic energy of such an electron depends on the total number of collisions. It is also possible that electron makes several collisions inside the metal and loses so much energy that it fails to come out. So the kinetic energy of the photoelectrons coming out may be anything between zero & $(h\nu - \phi)$. Thus, it can be written

$$K_{\max} = h\nu - \phi \quad \text{--- (II)}$$

This is called Einstein's photoelectric equation.

If m be the mass of electron & $v_{\max}$ be the velocity, then

$$K_{\max} = \frac{1}{2} m v_{\max}^2$$

Therefore equation (II) can be written as

$$\frac{1}{2} m v_{\max}^2 = h\nu - \phi \quad \text{--- (III)}$$

If the frequency of the incident light is decreased kinetic energy decreases.

For a particular frequency ν_0 called the threshold frequency, $K_{\max} = 0$

Therefore from equation (II), we have

$$h\nu_0 = \phi \quad \text{--- (IV)}$$

Substituting this value of ϕ in equation (II), we have

$$K_{\max} = h\nu - h\nu_0$$

$$\text{or } K_{\max} = h(\nu - \nu_0) \quad \text{--- V}$$

This is Einstein's photo-electric equation.

Explanation of Laws :-

1. The intensity of the light does not appear in the equation (II). Doubling the intensity of light means that twice as many photons strike the surface and twice as many photoelectrons are released. Thus the photo-current is directly proportional to the intensity of the incident light. Here only photons is doubled not the energy. The energy of each photon remains the same. Thus kinetic energy remains the same. Thus law (1) and (2) are explained.
2. It is clear from equation (II) that kinetic energy is a function of frequency only. It increases linearly with the increase in the frequency of the incident light. This is photoelectric law (3).
3. It is clear from equation (II) that if the frequency of the incident light ν is less than the threshold frequency ν_0 , kinetic energy of the photo-electrons would become negative which is impossible. Hence no photoelectron is ejected if $\nu < \nu_0$. This is photo-electric law (4).
4. Photon is a concentrated bundle of energy. Therefore, when it strikes the electron, whole of the energy is instantaneously supplied to the electron. It is not spread over a large area as in the case of wave theory. This is why without any time lag photoelectron is ejected. This is photo-electric law (5).

Thus Einstein's equation admirably explains all the facts about photo-electric effect.

Comparison Of Compton Effect and Photoelectric Effect

Let us consider a Compton collision between a photon and a free electron (initially at rest). The photon has energy $h\nu$ and momentum $h\nu/c$. Suppose the photon transfers entire energy (and momentum) to the electron which acquires a velocity v . Then from consideration of conservation of energy

$$h\nu = \frac{1}{2}mv^2 \quad \text{--- (1)}$$

from conservation of momentum

$$\frac{h\nu}{c} = mv \quad \text{--- (2)}$$

Here $m = m_{\text{rest}}$ of electron.

From (1)/(ii) we have

$$\frac{h\nu \times c}{h\nu} = \frac{\frac{1}{2}mv^2}{mv}$$

$$\text{or } \boxed{v = 2c} \quad \text{--- (3)}$$

This is impossible according to the relativistic mechanics. Hence we conclude that a 'free' electron cannot absorb all the energy of a photon and conserve both energy and momentum.

In photoelectric effect, an electron absorbs the incident photon completely before it leaves the photon sensitive surface as a photoelectron. Obviously, this electron cannot be free, that is it must be bound to an atom otherwise it could not absorb the photon completely. Hence, we conclude that the photoelectric effect cannot occur for completely free electrons; the electrons must be bound to an atom.

Thus, there is an important difference between the photoelectric effect and the Compton Effect. In the photoelectric effect, the incident photon completely disappears and all of its energy is given to the photoelectron. In Compton effect, there is still a photon after the collision but its energy is less than that of the incident photon, only a part of the energy of the incident photon is transferred to an electron in the scatter. Thus, in Compton Effect, the electron can be free.